

WASHINGTON DULLES, VA, USA

WMO: 724030

Lat: 38.9767N

Lon: 77.4858W

Elev: 290

StdP: 14.54

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93738

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	16.6	-3.6	4.6	16.1	1.2	5.9	20.7	27.2	31.9	25.0	31.8	7.1	310	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	93.2	74.5	90.7	73.7	88.3	72.7	77.5	88.4	76.3	86.1	75.1	84.0	8.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.5	130.2	81.3	73.4	125.4	80.3	72.3	121.0	79.4	41.0	88.6	39.7	86.2	38.6	84.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.1	16.9	DB	4.4	96.9	5.3	3.6	0.6	99.4	-2.6	101.5	-5.5	103.5	-9.4	106.1
			WB	3.3	80.0	5.0	1.8	-0.3	81.3	-3.2	82.4	-6.0	83.4	-9.7	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	33.8	36.5	44.5	55.4	64.1	72.8	76.9	75.7	68.9	57.0	46.1	38.1
	DBStd	17.19	10.52	9.69	9.69	8.75	7.79	5.90	4.81	4.92	6.70	8.19	8.53	8.86
	HDD50	1763	515	391	230	41	2	0	0	0	0	24	174	386
	HDD65	4557	968	799	640	310	116	10	0	0	37	273	569	835
	CDD50	3921	12	13	57	203	439	683	835	798	566	242	56	17
	CDD65	1238	0	0	3	22	88	243	370	333	152	26	1	0
	CDH74	11523	0	2	37	286	842	2196	3626	3058	1244	223	8	1
	CDH80	4322	0	0	7	76	254	793	1529	1208	408	47	0	0
Wind	WSAvg	6.8	7.8	8.0	8.6	8.1	6.6	6.1	5.6	5.4	5.8	6.2	6.8	6.8
Precipitation	PrecAvg	41.0	2.7	2.7	3.4	3.1	4.0	4.0	3.5	3.9	3.7	3.1	3.3	3.2
	PrecMax	65.7	6.6	6.3	7.6	7.4	10.3	18.2	7.2	10.7	11.3	9.2	7.8	6.7
	PrecMin	28.9	0.4	0.2	1.0	0.3	0.8	0.5	0.9	0.4	0.6	0.0	0.2	0.4
	PrecStd	8.0	1.5	1.6	1.4	1.6	2.1	3.1	1.7	2.2	2.6	2.2	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.3	70.2	78.8	87.3	89.6	94.0	97.1	95.6	92.2	85.0	74.6	69.0
		MCWB	59.9	57.8	60.9	67.0	71.3	74.9	76.3	76.2	71.9	69.6	61.9	60.9
	2%	DB	61.8	62.8	72.0	81.7	86.2	90.3	93.5	92.6	88.7	79.9	69.2	62.5
		MCWB	56.4	52.7	57.6	63.4	70.3	74.0	75.0	74.3	71.3	67.0	59.6	57.3
	5%	DB	56.1	56.6	66.3	76.6	82.8	87.7	90.9	89.5	84.8	75.5	64.9	56.9
		MCWB	50.4	47.8	54.8	61.3	68.4	72.5	74.4	73.3	69.7	64.6	57.3	51.1
	10%	DB	49.0	51.4	61.3	71.8	78.9	84.8	88.1	86.7	81.2	71.5	60.9	51.8
		MCWB	43.4	44.5	51.6	59.0	66.6	71.2	73.5	72.0	68.4	63.3	53.6	46.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	60.8	63.3	69.4	74.8	78.1	79.5	78.6	76.5	73.5	65.5	63.2
		MCDB	64.8	68.4	74.3	80.4	85.7	89.3	91.8	90.2	85.2	79.0	69.8	66.7
	2%	WB	57.0	54.4	60.0	66.4	72.4	76.1	77.7	77.1	74.6	70.3	62.3	58.1
		MCDB	61.8	60.0	68.9	77.0	82.5	86.6	88.8	87.9	82.1	76.6	67.4	62.2
	5%	WB	50.5	49.5	56.5	63.4	70.4	74.7	76.4	75.8	73.0	67.6	58.7	52.5
		MCDB	55.2	55.5	64.5	73.6	79.4	84.1	86.5	84.8	79.9	73.8	63.4	56.0
	10%	WB	44.2	44.9	52.5	60.5	68.4	73.2	75.2	74.5	71.3	64.0	54.8	46.6
		MCDB	48.3	50.8	60.4	70.1	76.4	81.5	84.3	82.6	77.6	69.9	59.7	50.8
Mean Daily Temperature Range	5% DB	MDBR	16.2	17.9	20.2	22.2	20.6	19.6	19.4	19.6	19.8	20.8	19.6	16.5
		MCDBR	23.6	26.2	28.5	29.4	25.2	22.7	22.8	22.5	24.1	25.8	25.8	22.8
		MCWBR	18.6	17.8	17.2	14.6	12.0	9.3	8.0	8.0	9.8	13.7	16.9	17.5
	5% WB	MCDBR	21.2	22.3	25.1	24.4	21.6	19.9	19.3	19.1	17.8	20.4	19.7	19.7
		MCWBR	18.2	17.2	17.1	13.6	10.9	9.1	8.0	8.0	8.7	12.3	15.4	17.3
Clear-Sky Solar Irradiance	taub	0.305	0.322	0.356	0.402	0.441	0.504	0.520	0.518	0.432	0.362	0.322	0.308	
	taud	2.470	2.403	2.376	2.273	2.192	2.055	2.031	2.015	2.235	2.432	2.502	2.498	
	Ebn at Noon	277	286	286	278	268	250	245	241	257	268	268	267	
	Edh at Noon	25	31	35	42	46	53	54	53	40	30	24	23	
All-Sky Solar Radiation	RadAvg	647	917	1218	1538	1730	1910	1873	1672	1367	1024	740	569	
	RadStd	57	104	80	118	183	145	102	84	169	115	62	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	-180	+266	+148
(47)	Regional (64 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+122

Nomenclature: See separate page